ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Assignment 7 - Due date 03/19/26

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github. And to do so you will need to fork our repository and link it to your RStudio.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A07_Sp26.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Packages needed for this assignment: "forecast","tseries". Do not forget to load them before running your script, since they are NOT default packages.\

Set up

```
#Load/install required package here
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(Kendall)
library(tseries)
library(outliers)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr   1.1.4     v stringr 1.5.1
## v forcats 1.0.0     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

Importing and processing the data set

Consider the data from the file "Net_generation_United_States_all_sectors_monthly.csv". The data corresponds to the monthly net generation from January 2001 to December 2020 by source and is provided by the US Energy Information and Administration. **You will work with the natural gas column only**.

Q1

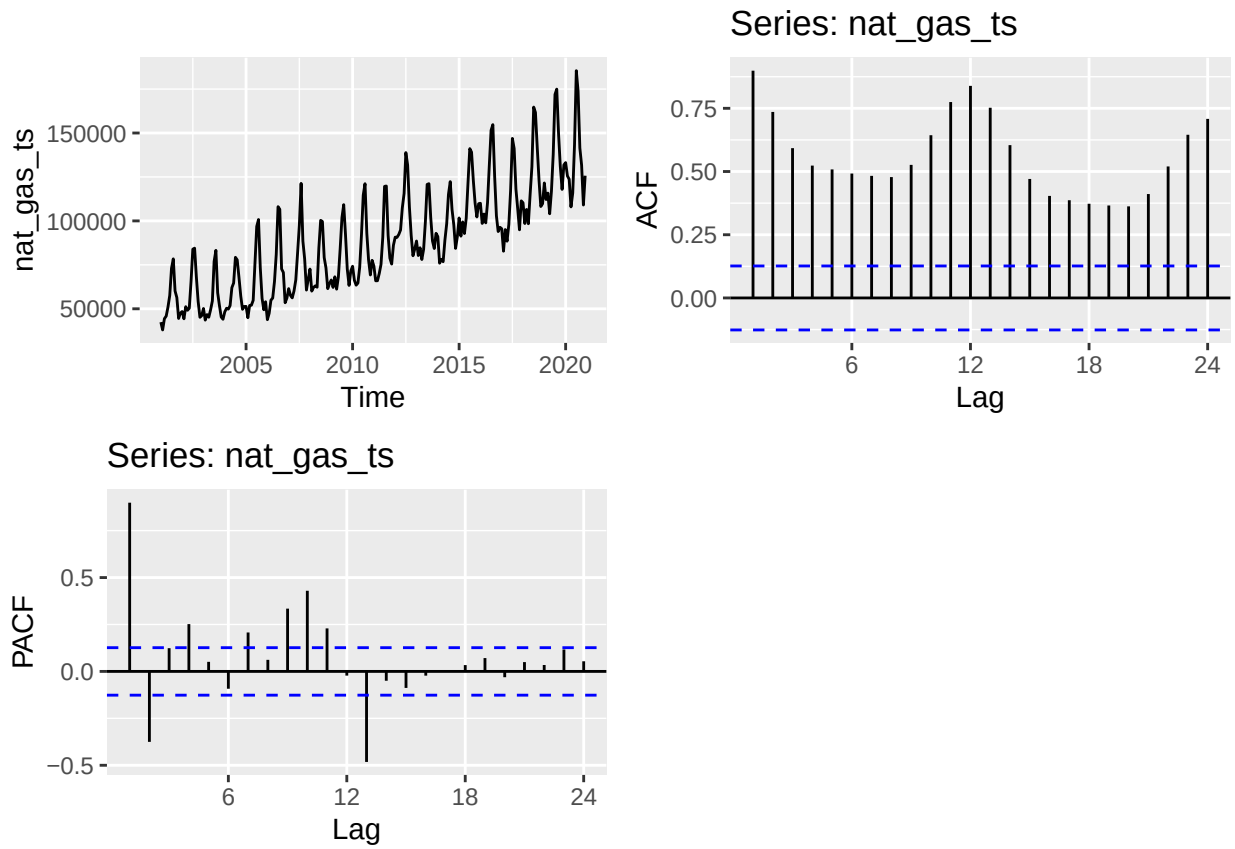
Import the csv file and create a time series object for natural gas. Make you sure you specify the **start=** and **frequency=** arguments. Plot the time series over time, ACF and PACF.

```r
net_gen <- read.csv(file="./Data/Net_generation_United_States_all_sectors_monthly.csv",header=TRUE,skip=

 #Inspect data
head(net_gen)
```

```
##      Month all.fuels..utility.scale..thousand.megawatthours
## 1 Dec 2020                                          344970.4
## 2 Nov 2020                                          302701.8
## 3 Oct 2020                                          313910.0
## 4 Sep 2020                                          334270.1
## 5 Aug 2020                                          399504.2
## 6 Jul 2020                                          414242.5
##   coal.thousand.megawatthours natural.gas.thousand.megawatthours
## 1                    78700.33                           125703.7
```

```
## 2                    61332.26                          109037.2
## 3                    59894.57                          131658.2
## 4                    68448.00                          141452.7
## 5                    91252.48                          173926.6
## 6                    89831.36                          185444.8
##   nuclear.thousand.megawatthours
## 1                       69870.98
## 2                       61759.98
## 3                       59362.46
## 4                       65727.32
## 5                       68982.19
## 6                       69385.44
##   conventional.hydroelectric.thousand.megawatthours
## 1                                          23086.37
## 2                                          21831.88
## 3                                          18320.72
## 4                                          19161.97
## 5                                          24081.57
## 6                                          27675.94
```

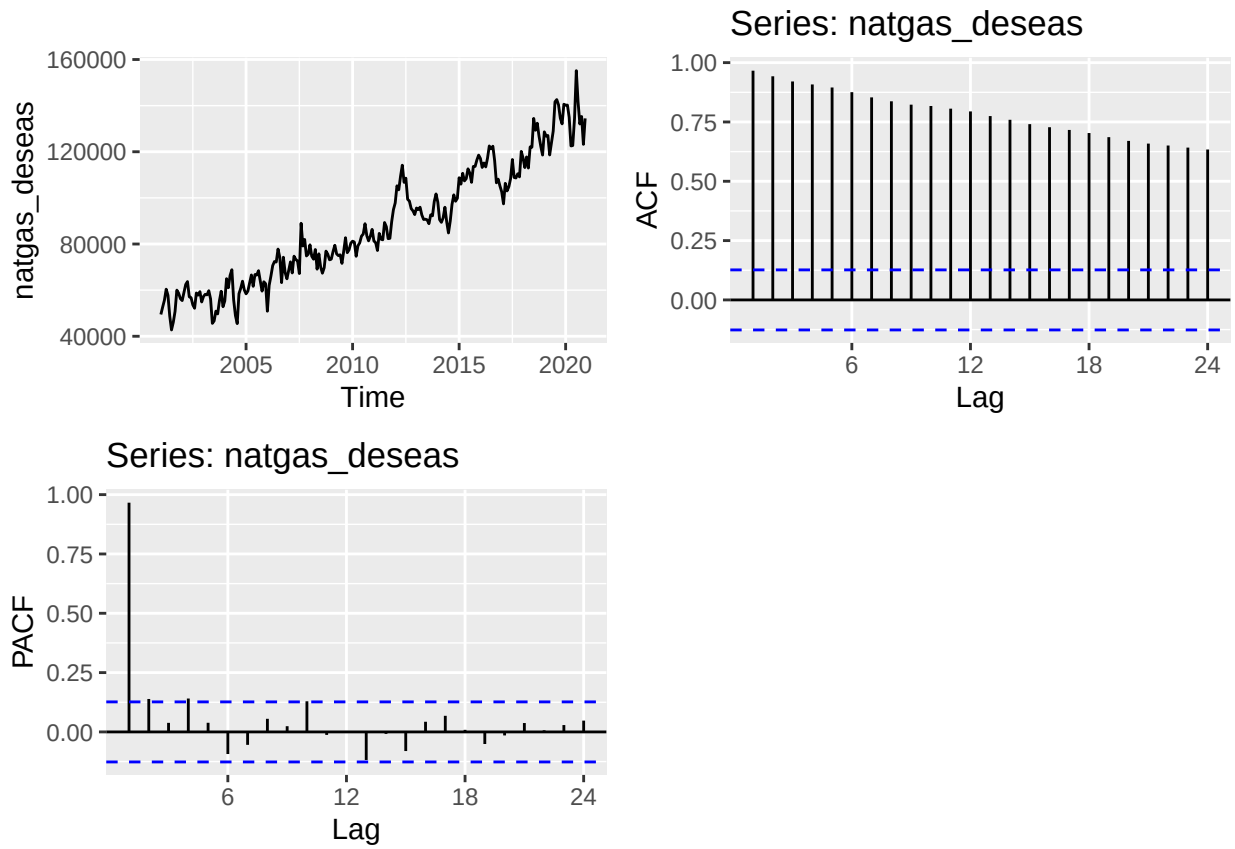
```r
nvar <- ncol(net_gen) - 1
nobs <- nrow(net_gen)

#Preparing the data - create date object and rename columns
nat_gas_df <-net_gen %>%
  mutate( Month = my(Month) ) %>%
  rename( Nat_Gas = natural.gas.thousand.megawatthours ) %>%
  arrange( Month )%>%
  select(Month, Nat_Gas)


nat_gas_ts <- ts(nat_gas_df$Nat_Gas,
               frequency = 12,
               start = c(2001, 1))


ts_plot <- autoplot(nat_gas_ts)
ts_acf <- autoplot(Acf(nat_gas_ts, plot= FALSE))
ts_pacf <- autoplot(Pacf(nat_gas_ts, plot= FALSE))

plot_grid(ts_plot, ts_acf, ts_pacf)
```



Q2

Using the *decompose*() and the *seasadj*() functions create a series without the seasonal component, i.e., a deseasonalized natural gas series. Plot the deseasonalized series over time and corresponding ACF and PACF. Compare with the plots obtained in Q1.

```
natgas_decomp <- decompose(nat_gas_ts, type = "additive")
natgas_deseas <- seasadj(natgas_decomp)

ts_plot <- autoplot(natgas_deseas)
ts_acf <- autoplot(Acf(natgas_deseas, plot= FALSE))
ts_pacf <- autoplot(Pacf(natgas_deseas, plot= FALSE))

plot_grid(ts_plot, ts_acf, ts_pacf)
```



Modeling the seasonally adjusted (i.e., the deseasonalized series)

Q3

Run the ADF test and Mann Kendall test on the deseasonalized data from Q2. Report and explain the results.

```
print((adf.test(natgas_deseas,alternative="stationary")))
```

```
## Warning in adf.test(natgas_deseas, alternative = "stationary"): p-value smaller
## than printed p-value
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  natgas_deseas
## Dickey-Fuller = -4.0271, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```
print(MannKendall(natgas_deseas))
```

```
## tau = 0.843, 2-sided pvalue =< 2.22e-16
```

Answer: The significant p-value ($p < 0.01$) from the ADF test indicates that the deseasonalized natural gas data is stationary. The Mann-Kendall test indicates the series has a strong and significant positive trend (tau = 0.843, $p < 0.001$). This means that even though the seasonality has been removed, there is still a clear upward trend in the data.

Q4

Using the plots from Q2 and test results from Q3 identify the ARIMA model parameters p, d and q. Note that in this case because you removed the seasonal component prior to identifying the model you don't need to worry about seasonal component. Clearly state your criteria and any additional function in R you might use. DO NOT use the *auto.arima*() function. You will be evaluated on ability to understand the ACF/PACF plots and interpret the test results.

```
Model_100 <- Arima(natgas_deseas,order=c(1,0,0),include.drift=TRUE)
print(Model_100)
```

```
## Series: natgas_deseas
## ARIMA(1,0,0) with drift
##
## Coefficients:
##          ar1  intercept     drift
##       0.7182  44800.489  359.3965
## s.e.  0.0445   2295.974   16.4299
##
## sigma^2 = 26630969:  log likelihood = -2391.11
## AIC=4790.22   AICc=4790.39   BIC=4804.14
```

Because the series is stationary, as indicated by the ADF test, the series should not be differenced ($d = 0$). In the plots from Q2, the ACF has a slow decay, while the PACF cuts off after one spike. These patterns in the ACF and PACF indicate that there is a autoregressive component in the series. The PACF cutting off after one spike indicates that the order of the AR component is 1 ($p = 1$).

Q5

Use `Arima()` from package "forecast" to fit an ARIMA model to your series considering the order estimated in Q4. You should allow constants in the model, i.e., `include.mean = TRUE` or `include.drift=TRUE`. **Print the coefficients** in your report. Hint: use the `cat()` or `print()` function to print.

```
Model_100 <- Arima(natgas_deseas,order=c(1,0,0),include.drift=TRUE)
print(Model_100)
```

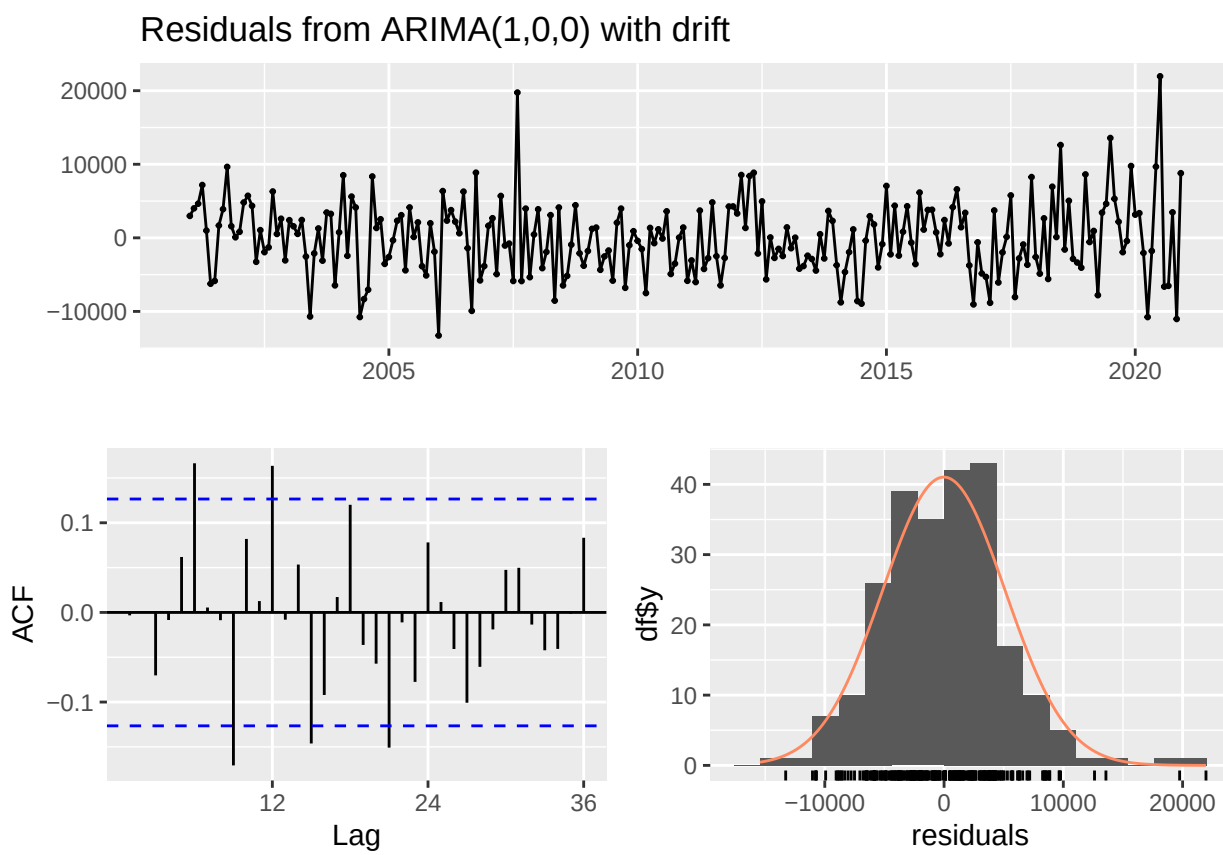
```
## Series: natgas_deseas
## ARIMA(1,0,0) with drift
##
## Coefficients:
##          ar1  intercept     drift
##       0.7182  44800.489  359.3965
## s.e.  0.0445   2295.974   16.4299
##
## sigma^2 = 26630969:  log likelihood = -2391.11
## AIC=4790.22   AICc=4790.39   BIC=4804.14
```

Q6

Now plot the residuals of the ARIMA fit from Q5 along with residuals ACF on the same window. You may use the *checkresiduals*() function to get the plots. Do the residual series look like a white noise series? Why?

Modeling the original series (with seasonality)

```
checkresiduals(Model_100)
```



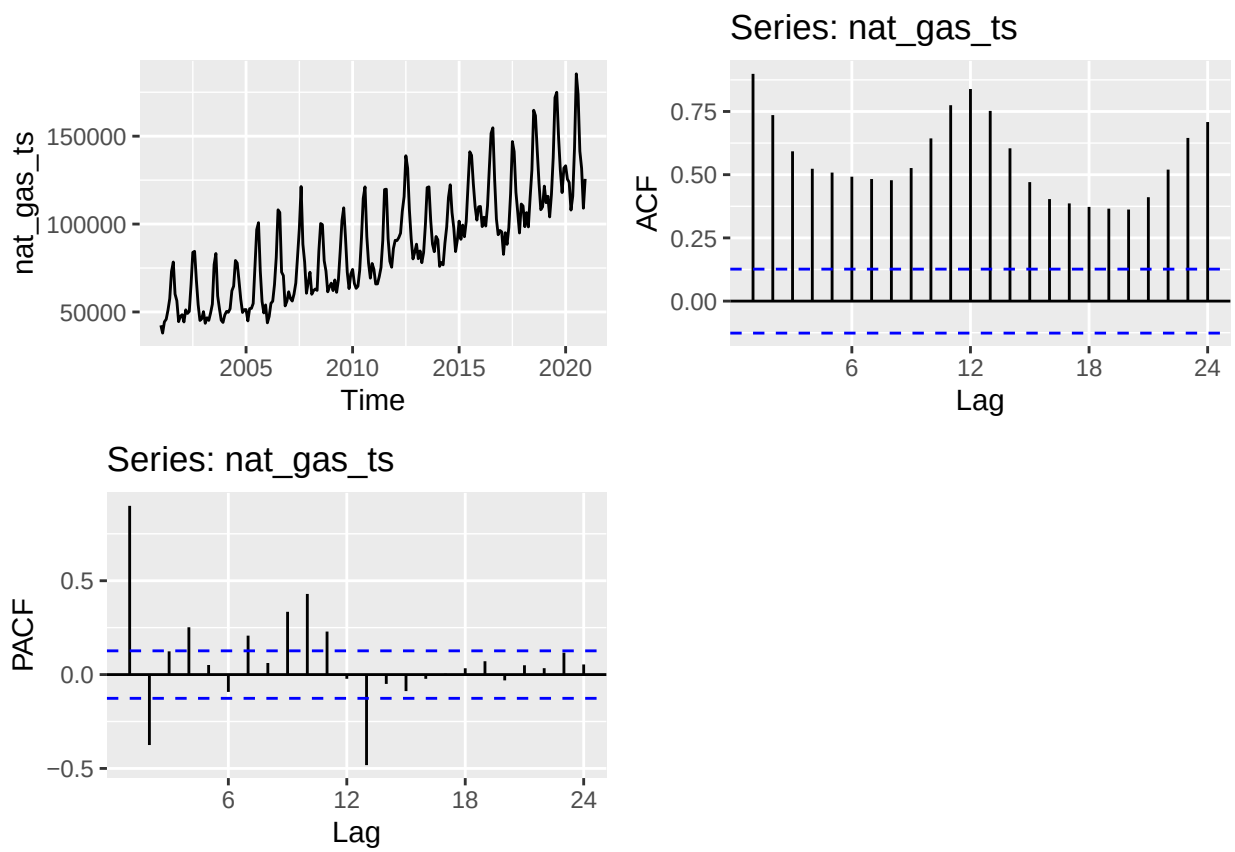

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with drift
## Q* = 47.775, df = 23, p-value = 0.001787
##
## Model df: 1.   Total lags used: 24
```

Answer: While the residuals appear to generally resemble a white noise series, the results of the Ljung-box test show otherwise. The significant p-value for the Ljung-box test (p-value = 0.001787) indicates that there is still autocorrelation in the residuals and the model does not capture the structure of the data.

Q7

Repeat Q3-Q6 for the original series (the complete series that has the seasonal component). Note that when you model the seasonal series, you need to specify the seasonal part of the ARIMA model as well, i.e., P, D and Q.

```
ts_plot_og <- autoplot(nat_gas_ts)
ts_acf_og <- autoplot(Acf(nat_gas_ts, plot= FALSE))
ts_pacf_og <- autoplot(Pacf(nat_gas_ts, plot= FALSE))

plot_grid(ts_plot_og, ts_acf_og, ts_pacf_og)
```




```
print((adf.test(nat_gas_ts,alternative="stationary")))
```

```
## Warning in adf.test(nat_gas_ts, alternative = "stationary"): p-value smaller
## than printed p-value

##
##  Augmented Dickey-Fuller Test
##
## data:  nat_gas_ts
## Dickey-Fuller = -8.9602, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```r
print(MannKendall(nat_gas_ts))
```

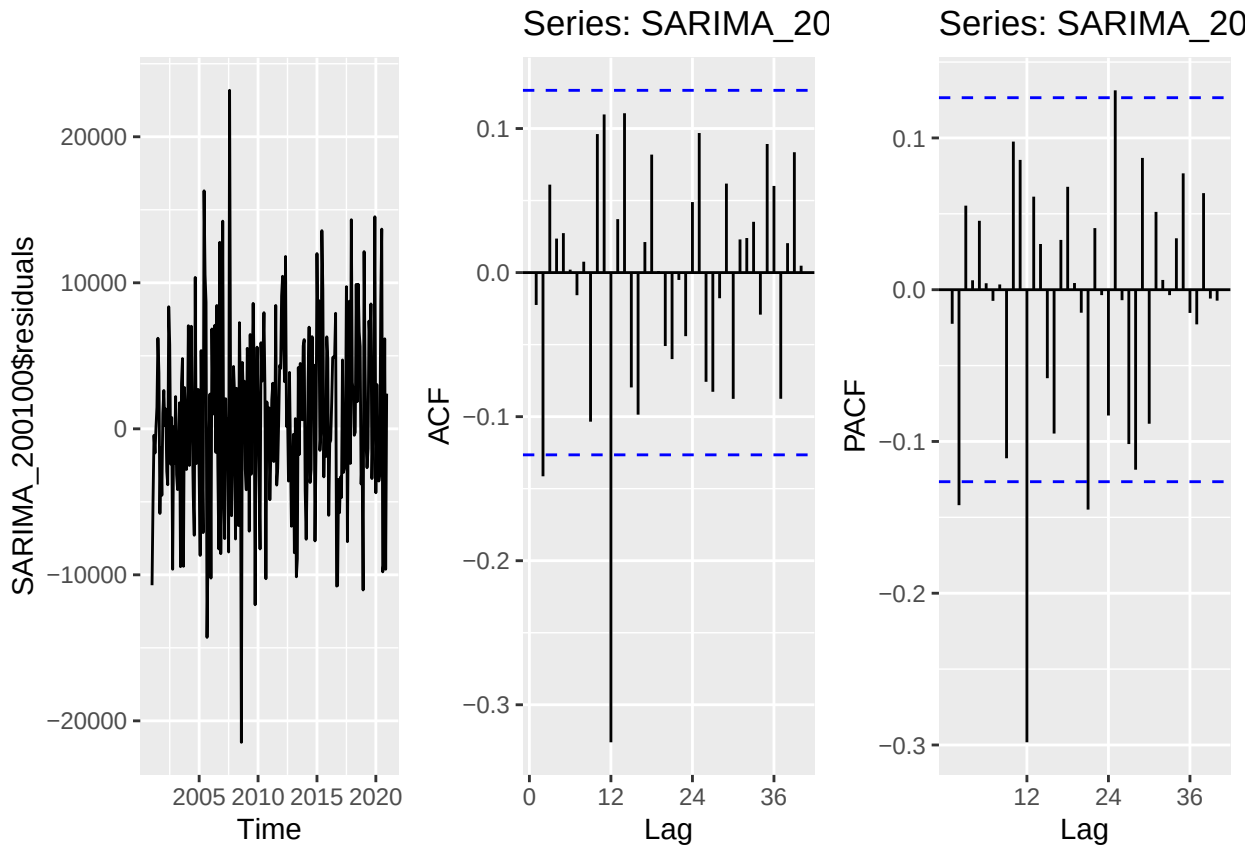
```
## tau = 0.651, 2-sided pvalue =< 2.22e-16
```

```r
#the original series has the same

SARIMA_200100 <- Arima(nat_gas_ts,
                       order=c(2,0,0),
                       seasonal=c(1,0,0),
                       include.drift=FALSE)
print(SARIMA_200100)
```

```
## Series: nat_gas_ts
## ARIMA(2,0,0)(1,0,0)[12] with non-zero mean
##
## Coefficients:
##          ar1     ar2    sar1      mean
##       0.7671  0.0561  0.9025  89680.57
## s.e.  0.0670  0.0655  0.0264  16827.68
##
## sigma^2 = 40260998:  log likelihood = -2450.6
## AIC=4911.19   AICc=4911.45   BIC=4928.6
```

```r
plot_grid(
  autoplot(SARIMA_200100$residuals),
  autoplot(Acf(SARIMA_200100$residuals,lag.max=40,plot=FALSE),series="Residuals"),
  autoplot(Pacf(SARIMA_200100$residuals,lag.max=40,plot=FALSE),series="Residuals"),
  nrow=1,ncol=3
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `series`

## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown
## parameters: `series`
```

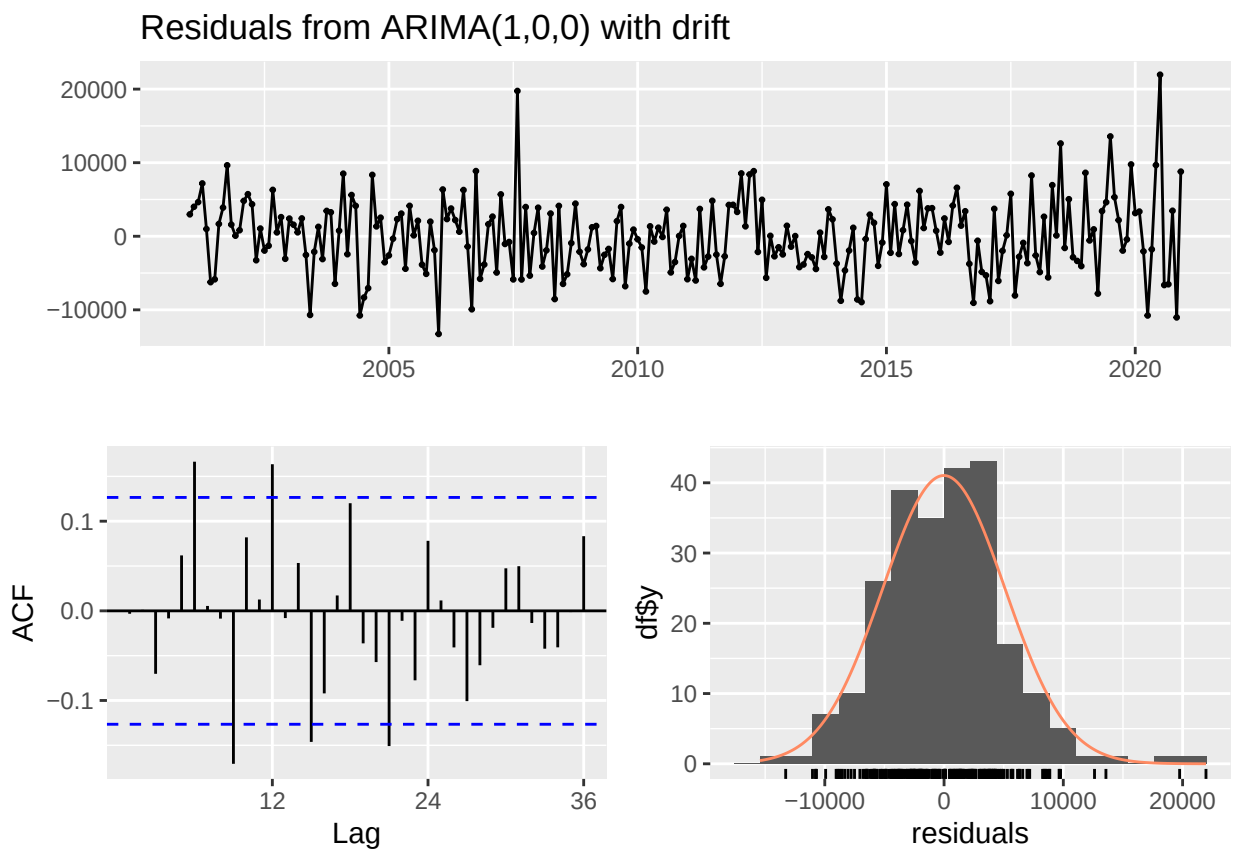


Answer: Because the series is stationary, as indicated by the ADF test, differencing was not needed (d = 0). The ACF shows strong autocorrelation and seasonality, while the PACF has two significant spikes before tapering off, indicating an autoregressive component of order p = 2. Additionally, the single spike at lag 12 in the ACF suggests a seasonal component, so a seasonal AR term was included (P = 1). When assessing the residuals of the SARIMA, the residuals appear fairly random, with potential slight seasonality. The ACF and PACF of the residuals support that seasonality was not fully removed from the series, as each had a strong negative spike at lag 12.

Q8

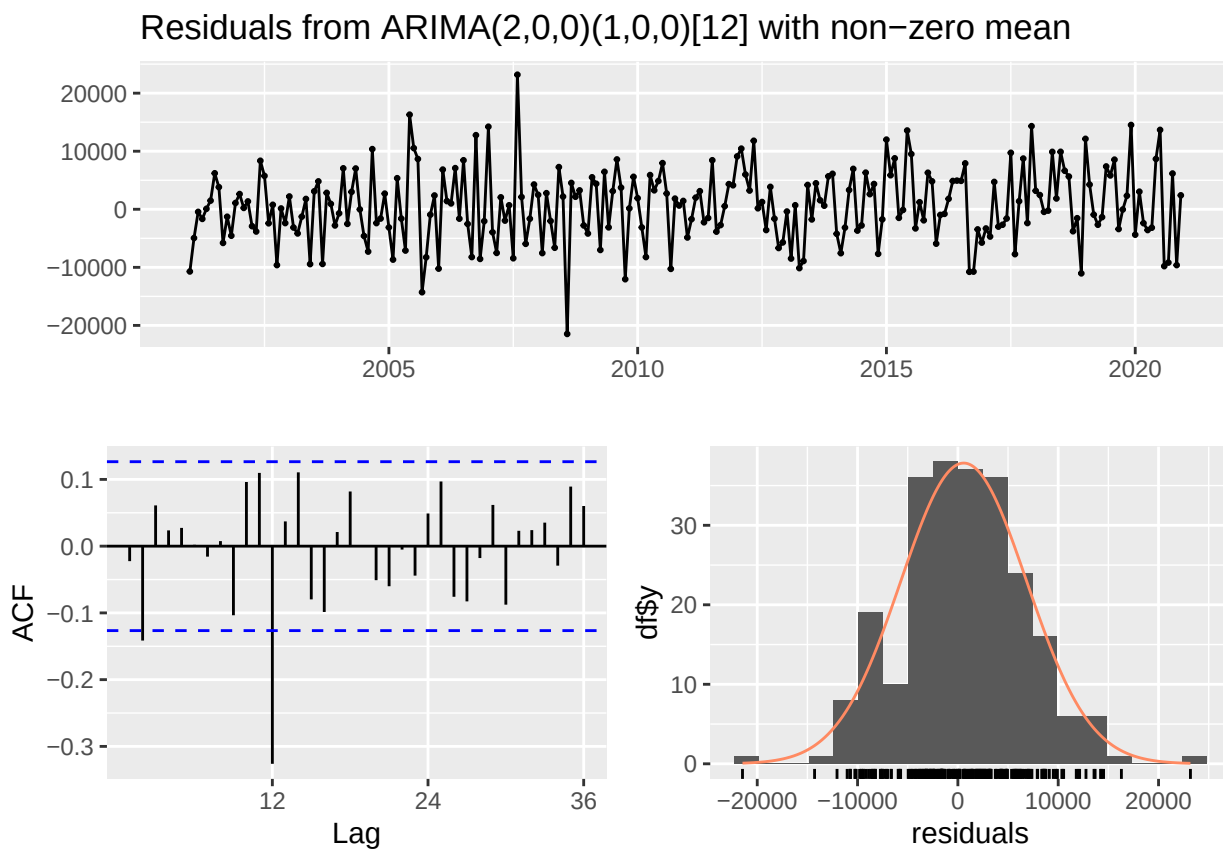
Compare the residual series for Q7 and Q6. Can you tell which ARIMA model is better representing the Natural Gas Series? Is that a fair comparison? Explain your response.

```
checkresiduals(Model_100)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with drift
## Q* = 47.775, df = 23, p-value = 0.001787
##
## Model df: 1.   Total lags used: 24
```

```r
checkresiduals(SARIMA_200100)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,0,0)(1,0,0)[12] with non-zero mean
## Q* = 53.823, df = 21, p-value = 0.0001047
##
## Model df: 3.   Total lags used: 24
```

Answer: The SARIMA model appears to better represent the natural gas series because it explicitly accounts for seasonality, whereas the ARIMA model was fit to a deseasonalized series. However, this is not a completely fair comparison since the models were fit to different versions of the data. The SARIMA model captures both seasonal and non-seasonal patterns, making it more appropriate for the original dataset.

Checking your model with the auto.arima()

Please do not change your answers for Q4 and Q7 after you ran the *auto.arima*(). It is **ok** if you didn't get all orders correctly. You will not loose points for not having the same order as the *auto.arima*().

Q9

Use the *auto.arima*() command on the **deseasonalized series** to let R choose the model parameter for you. What's the order of the best ARIMA model? Does it match what you specified in Q4?

```r
print(auto.arima(natgas_deseas))
```

```
## Series: natgas_deseas
## ARIMA(1,1,1) with drift
##
## Coefficients:
##          ar1      ma1     drift
##       0.7065  -0.9795  359.5052
## s.e.  0.0633   0.0326   29.5277
##
## sigma^2 = 26980609:  log likelihood = -2383.11
## AIC=4774.21   AICc=4774.38   BIC=4788.12
```

The auto.arima results do not match the order I specified in Q4. In Q4 my model was ARIMA(1,0,0), however the order of the autofit model is ARIMA(1,1,1).

Q10

Use the *auto.arima*() command on the **original series** to let R choose the model parameters for you. Does it match what you specified in Q7?

```r
print(auto.arima(nat_gas_ts))
```

```
## Series: nat_gas_ts
## ARIMA(1,0,0)(0,1,1)[12] with drift
##
## Coefficients:
##          ar1     sma1     drift
##       0.7416  -0.7026  358.7988
## s.e.  0.0442   0.0557   37.5875
##
## sigma^2 = 27569124:  log likelihood = -2279.54
## AIC=4567.08   AICc=4567.26   BIC=4580.8
```

Answer: The auto.arima results do not match the order I specified in Q7. In Q7 my model was ARIMA(2,0,0)(1,0,0), however the order of the autofit model is ARIMA(1,0,0)(0,1,1).